WORKED SOLUTION

STEP I 2012, Question 11

Mechanics • Pulleys and Friction

STEP I 2012, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q11

The key idea

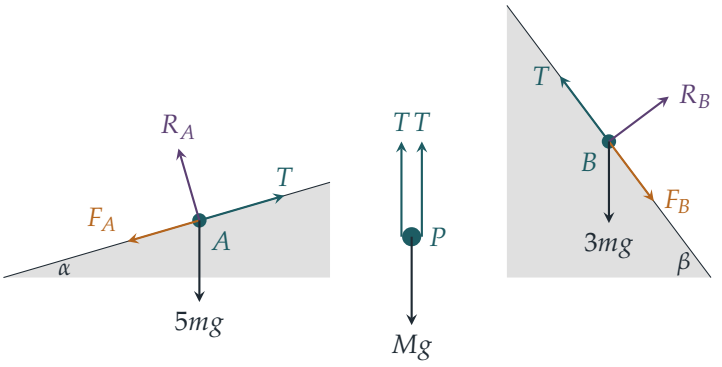
Because every pulley is smooth and the string is light, the tension is the same all the way along it. So each particle gives one equation linking the tension to friction, and the pulley P gives a third. The angles are chosen from the $7, 24, 25$ and $3, 4, 5$ triangles, so all the sines and cosines are exact fractions. For the moving case we need one more ingredient: how the accelerations of A, B and P are tied together by the fixed length of the string.

Solution

We model A, B as particles and the string as light and inextensible. Pulleys Q and R are smooth, light and fixed, and P is smooth with mass M. Let $\alpha = \arctan\frac{7}{24}$ and $\beta = \arctan\frac{4}{3}$. From the right-angled triangles with sides $7, 24, 25$ and $3, 4, 5$,

$$\sin\alpha = \tfrac{7}{25}, \quad \cos\alpha = \tfrac{24}{25}, \qquad \sin\beta = \tfrac{4}{5}, \quad \cos\beta = \tfrac{3}{5}.$$

Let T be the tension. Since all the pulleys are smooth, T is the same in every segment.



In both parts the particles are on the point of moving, or moving, *up* their planes, so

friction acts *down* each plane, as drawn. Resolving perpendicular to the planes,

$$R_A = 5mg\cos\alpha = \tfrac{24}{5}mg, \qquad R_B = 3mg\cos\beta = \tfrac{9}{5}mg.$$

Part (i)

Both particles are on the point of slipping, so friction is limiting: $F_A = \mu R_A$ and $F_B = \mu R_B$. Resolving along each plane (equilibrium):

$$\begin{aligned} A: &\quad T = 5mg\sin\alpha + \mu R_A = \tfrac{mg}{5}(7+24\mu),\\ B: &\quad T = 3mg\sin\beta + \mu R_B = \tfrac{mg}{5}(12+9\mu). \end{aligned}$$

Equating, $7+24\mu = 12+9\mu$, so

$$\mu = \tfrac{1}{3},$$

and then $T = \frac{mg}{5}(7+8) = 3mg$.

The pulley P is in equilibrium with two vertical string segments pulling it up and its weight pulling it down, so

$$2T = Mg \quad\Longrightarrow\quad M = \frac{2\cdot 3mg}{g} = 6m,$$

as required.

Part (ii)

Now $M = 9m$, which is heavier than the $6m$ that could just be held in limiting equilibrium, so we expect P to descend and drag both particles up their planes.

Ask yourself How are the three accelerations related?

Let A and B have accelerations a and b up their planes, and let P have acceleration p downwards. The string length $AQ + QP + PR + RB$ is fixed. If A moves a distance x up its plane and B a distance y up its plane, then AQ and RB shorten by x and y; if P drops by z, then QP and PR each lengthen by z. Hence $2z = x + y$ at all times, and differentiating twice,

$$2p = a + b.$$

With $\mu = \frac{1}{3}$, the friction while sliding up is $\frac{1}{3}R_A = \frac{8}{5}mg$ on A and $\frac{1}{3}R_B = \frac{3}{5}mg$ on B. Newton's second law along each plane and vertically for P gives

$$\begin{aligned} A: &\quad T - \tfrac{7}{5}mg - \tfrac{8}{5}mg = 5ma, && \text{so } T - 3mg = 5ma,\\ B: &\quad T - \tfrac{12}{5}mg - \tfrac{3}{5}mg = 3mb, && \text{so } T - 3mg = 3mb,\\ P: &\quad 9mg - 2T = 9mp. \end{aligned}$$

Put $T = 3mg + X$. Then $a = \dfrac{X}{5m}$, $b = \dfrac{X}{3m}$, and the string condition gives

$$p = \tfrac{1}{2}(a+b) = \frac{4X}{15m}.$$

Substituting into the equation for P:

$$3mg - 2X = \frac{12X}{5} \quad \Longrightarrow \quad X = \frac{15mg}{22}.$$

Therefore

$$a = \frac{3g}{22} \text{ up the plane}, \qquad b = \frac{5g}{22} \text{ up the plane}, \qquad p = \frac{2g}{11} \text{ vertically downwards},$$

with tension $T = \frac{81}{22}mg$.

Ask yourself How do we know both particles really do move up, rather than one staying put?

From the equations above, a particle slides up its plane only if $T > 3mg$, and stays at rest (or would slide down) only if $T \leq 3mg$. The value $3mg$ is the same for both particles, so either both move up or neither does. If neither moved up, then $a + b \leq 0$, so $p \leq 0$; but $T \leq 3mg$ gives $9mg - 2T \geq 3mg > 0$, which forces $p > 0$. That is a contradiction, so both move up, and our solution (with $a, b > 0$) is consistent.

Checks

- In part (i), substituting $\mu = \frac{1}{3}$ into both particle equations gives the same tension, $3mg$.
- In part (ii), $2p = \frac{4g}{11} = \frac{3g}{22} + \frac{5g}{22} = a + b$, as the string requires, and $T = \frac{81}{22}mg \approx 3.68mg$ lies between $3mg$ (the threshold for slipping) and $4.5mg$ (the value at which P would have no acceleration).
- A numerical simulation stepping the three equations of motion keeps the string length fixed up to rounding error, and $p \approx 0.182g$.
- If M were $6m$ again, the same method gives $X = 0$, so all accelerations vanish, matching part (i).

Common slips

- Taking $a = b = p$: the pulley P moves at half the combined rate at which the two particles move, because two segments of string share its motion.
- Putting friction up the plane: both particles are being pulled *up*, so friction opposes that and acts down the plane.
- Forgetting that both strings on P carry tension T, so the upward force on it is $2T$.

Beyond STEP

The relation $2p = a + b$ is a *constraint*: it cuts the number of independent coordinates from three to two. In Lagrangian mechanics, met in the first or second year, you pick

coordinates that satisfy the constraints automatically and never have to find the tension at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many pulley systems treated carefully through constraint equations, a good bridge to that viewpoint.

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